

SIDONG ZHANG

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🏠 <https://stonezhng.github.io>

Education

UMass Amherst's College of Information and Computer Sciences

Amherst, MA, USA

Doctor of Philosophy

Sep. 2020 - present

- Teaching assistant of graduate-level CS 651 optimization and CS 687 reinforcement learning
- Research assistant in Information Fusion lab

UMass Amherst's College of Information and Computer Sciences

Amherst, MA, USA

Master of Science

Sep. 2018 - Jan. 2021

- GPA: 3.82/4.0

Nanjing University, Software Institute

Nanjing, China

Bachelor of Engineering

Sep. 2014 - June 2018

- GPA: 4.35/5.0

First Author Work

Video2Reaction: Training Foundation Video Models to Predict Audience Reaction

Accepted to CVPR 2026 Workshop on Emerging Directions in Data for Multimodal Foundation Models Feb. 2026 - Apr. 2026

- Build based on Video2Reaction (V2R) we curated, a novel dataset of audience emotional reactions to movie clips, annotated with distributional emotion labels across 21 reaction categories
- Developed a cross-taxonomy emotion transfer pipeline that finetunes VLMs on V2R using word-token based scoring, enabling zero-shot transfer to a target emotion dataset (VCE) with up to +0.269 top-3 accuracy gain over baseline
- Demonstrated that V2R-pretrained VLMs (LLaVA-NeXT-Video-7B and Qwen2.5-VL-7B) match VideoMAE trained on the full 50K ED dataset using only 1% of target domain training data

Audio-Visual Speech Separation via Bottleneck Iterative Network

Accepted to ICML 2025 Workshop on Machine Learning for Audio

Feb. 2024 - Sep. 2024

- Addressed audio-visual speech separation on noisy mixtures from NTCD-TIMIT and LRS3
- Proposed a multimodal fusion framework combining audio and visual modalities via a bottleneck iterative architecture
- Achieved state-of-the-art SI-SDRi while reducing training time by 50% compared to prior SOTA
- Project webpage: <https://stonezhng.github.io/projects/avssbin/>

Understanding 3D Brain MRI Scans in Foundation Modeling

Ongoing work

Feb. 2026 -

- Existing foundation models (BiomedCLIP, MedGemma) use 2D medical image encoders, with 3D extensions largely treating volumes as sequences of 2D slices, losing inter-slice structural context
- Propose native 3D conv patch embeddings to better capture volumetric structure in brain MRIs, where information is equally distributed across all three axes
- Finetuning BiomedCLIP and MedSigLIP image encoders with 3D Conv patch embedding layers on brain MRI data
- Applying atlas-guided flexible patch sizing, assigning finer-grained patches to cognition-relevant regions (e.g., hippocampi) for cognitive state finetuning tasks

Longitudinal Multimodal Modeling for Alzheimer's Early Detection

Submitting to Medical Image Analysis

Sep. 2020 - July 2025

- Showed that task-agnostic representations learned from T1-weighted brain MRIs via mutual information maximization capture knowledge predictive of cognitive decline scores
- Trained a CNN from scratch and fine-tuned BiomedCLIP as two representation learning backbones
- Designed a training strategy for an existing AD forecasting model by partially freezing model parameters, stabilizing performance on small validation splits
- Evaluated using micro F1 on CN/MCI/AD classification and MCI-to-AD transition timing precision across 100 repeated experiments for statistical robustness
- Task-agnostic MRI representations from both CNN and BiomedCLIP improved 2-year forecasting F1 and transition timing accuracy

Second Author Work

Encoding Domain Insights into Multi-modal Fusion: Improved Performance at the Cost of Robustness

Accepted to ICML 2025 Workshop on Methods and Opportunities at Small Scale

Jan. 2025 – May 2025

- Compared fusion architectures with and without domain knowledge embedded in the model structure
- Evaluated on MMSD for sarcasm detection and a synthetic task with controllable domain knowledge levels; added Gaussian noise to inputs to probe robustness
- Found that aligning fusion design with domain priors improves clean-data accuracy under limited data but significantly degrades robustness to noisy inputs

Video2Reaction: Mapping Video to Audience Reaction in the Wild

Under review at ECCV 2026

Feb. 2025 - May 2025

- Curated a multimodal dataset mapping short movie clips to distributional emotional reactions of real-world viewers
- Collected movie clips and YouTube comments, extracting per-clip emotion distributions from viewer responses
- Deployed 3 LLMs to rephrase and independently label comment emotions; applied majority voting per comment and discarded samples with no consensus across all three models
- Established a comprehensive benchmark for distributional video-to-reaction modeling

Technical Skills

Languages: English (Proficient), Mandarin (Native)

Programming language: Java, Python, C

Research interests: multimodal learning, multimedia, healthcare, computer vision